

BUS 123 - Solving Business Problems with Technology

MATH-M10-L01

BUS 123 — MATH-M10-L01 — Foundations of Investing: Stocks, Bonds, Mutual Funds & ETFs

PRE-READING

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BUS123 · MATH-M10-L01 PRE-READING

FOUNDATIONS OF INVESTING: STOCKS, BONDS, MUTUAL FUNDS & ETFS

Meridian Advisory Group Case Study Gerrish School of Business · Endicott College · Fall 2026

CURRENT EVENTS HOOK

When a closely watched company prepares to enter the public markets, headlines often focus on its expected valuation and possible offering price. Investors still must ask a question that has no easy answer: *Is this security worth buying at that price?* The names and numbers change; the need to understand ownership, return, risk, and valuation does not.

That question — *how do I decide whether a security is worth owning?* — is exactly what this module is about. By the time you finish MATH-M10, you will be able to read a stock quote, calculate the return on a bond, and determine the net asset value of a mutual fund. Those are the building blocks of every serious investment decision.

CONNECT TO PRIOR KNOWLEDGE

In MATH-M09, you worked with the time value of money — specifically compound interest and present value. You learned that a dollar today is worth more than a dollar in the future because money can earn a return over time.

That exact principle drives everything in this lesson. When you buy a stock, you are paying today for a share of future profits. When you buy a bond, you are lending money today in exchange for interest payments and a return of principal in the future. When you invest in a mutual fund, you are pooling your present-day dollars with others to capture future market growth. The TVM logic does not disappear — it runs underneath every calculation you will do today.

PART 1 · STOCKS — OWNERSHIP IN A BUSINESS

What a Stock Actually Is

A **stock** is a share of ownership in a company. When Meridian Advisory Group recommends that a client buy 200 shares of a company, that client becomes a partial owner of the business — entitled to a share of its profits (paid as dividends) and a claim on its assets if the company is ever liquidated.

Companies issue stock to raise capital. Rather than taking out a bank loan, a business can sell pieces of itself to thousands of investors, using the proceeds to expand, hire, or pay down debt. In return, investors hope the company grows and that their shares become more valuable.

Common vs. Preferred Stock

There are two main types of stock, and they come with very different rights.

Common stock is what most people mean when they say "buying stock." Common shareholders have voting rights — they can vote on major corporate decisions such as who sits on the board of directors. However, they are last in line when dividends are distributed and last in line if the company goes bankrupt.

Preferred stock generally does not come with voting rights, but preferred shareholders receive declared dividends before common shareholders, and they have a higher claim than common shareholders on remaining assets if the company fails. With **cumulative preferred stock**, a skipped dividend accumulates as **dividends in arrears**. The arrears normally must be paid before common dividends resume, but they generally are not corporate debt unless the board has formally declared the dividend.

Key rule: Preferred shareholders are paid first. Common shareholders receive whatever is left.

Reading a Stock Quote

Every publicly traded company is identified by a **ticker symbol** — a short code used on exchanges for speed and precision. Coca-Cola is KO. Apple is AAPL. Meridian Advisory Group Holdings, in our case study, uses the symbol MAGH.

When you look up a stock quote, you will see several key data points:

Term	What It Means
Last Price	The most recent price at which a buyer and seller completed a trade
Bid Price	The highest price any buyer is currently willing to pay
Ask Price	The lowest price any seller is currently willing to accept
Bid-Ask Spread	Ask minus Bid — the immediate cost of executing a trade
52-Week High / Low	The highest and lowest traded prices over the past year
P/E Ratio	Price per share divided by earnings per share — how much investors pay per \$1 of earnings
Dividend Yield	Annual dividend divided by current price — income return expressed as a percentage

The **bid-ask spread** matters more than most beginners realize. Because you buy at the Ask and sell at the Bid, you are effectively at a small loss the moment a trade is executed. Understanding this gap is one mark of a disciplined investor.

Calculating Dividends

When a company declares dividends, preferred shareholders collect first. The calculation always follows the same steps:

1. Calculate total preferred dividends: **Rate × Par Value × Number of Preferred Shares**
2. Subtract from total dividends declared to find the **remainder for common shareholders**
3. Divide the remainder by the number of common shares to find the **dividend per common share**

Worked Example — Meridian Advisory Group

Meridian's model portfolio company has declared \$10,000 in dividends. The capital structure is: 8% preferred stock, \$50 par value, 1,000 shares; and 4,000 common shares outstanding.

- Preferred dividend per share: $8\% \times \$50 = \4.00
- Total preferred dividend: $\$4.00 \times 1,000 \text{ shares} = \$4,000$
- Remainder for common: $\$10,000 - \$4,000 = \$6,000$
- Common dividend per share: $\$6,000 \div 4,000 \text{ shares} = \1.50 per share

Calculating Return on Investment (ROI)

ROI measures how much you made (or lost) on an investment, expressed as a percentage of what you originally put in. It accounts for the price change, commissions, and any dividends received.

Formula:

$$\text{ROI} = (\text{Net Proceeds} - \text{Total Buy Cost} + \text{Dividends Received}) \div \text{Total Buy Cost}$$

Worked Example — Meridian Advisory Group

An analyst at Meridian bought 200 shares of MAGH at \$39.09 and sold them one year later at \$41.10. The commission rate was 1% on both transactions. MAGH paid a \$1.21 annual dividend per share.

Step	Calculation	Result
Buy cost (before commission)	$200 \times \$39.09$	\$7,818.00
Buy commission	$\$7,818 \times 1\%$	\$78.18
Total buy cost	$\$7,818 + \78.18	\$7,896.18
Sale proceeds (before commission)	$200 \times \$41.10$	\$8,220.00
Sell commission	$\$8,220 \times 1\%$	\$82.20
Net sale proceeds	$\$8,220 - \82.20	\$8,137.80
Total dividends received	$200 \times \$1.21$	\$242.00
ROI	$(\$8,137.80 - \$7,896.18 + \$242.00) \div \$7,896.18$	6.12%

PART 2 · BONDS — LENDING TO A COMPANY

What a Bond Actually Is

When you own a stock, you own a piece of the company. When you own a bond, you are a **creditor** — you have lent money to the company. The company promises to pay you interest on a schedule and return the full face amount (the **par value**, typically \$1,000) at a future date called the **maturity date**.

If a company goes bankrupt, bondholders are creditors and rank ahead of stockholders. Their exact priority depends on the bond's terms and other claims: secured creditors may rank ahead of unsecured bondholders. This creditor status generally makes a particular company's bonds less risky than its stock, though bonds still carry default, interest-rate, inflation, and liquidity risk.

How Bond Prices Work

Bond prices are always quoted as a **percent of par value**. This is the single most important rule in bond math, and the most common source of errors.

A bond quoted at **99.59** does not cost \$99.59. It costs **99.59% of \$1,000 = \$995.90**.

Quoted Price	Dollar Price	Status
Above 100 (e.g., 105.25)	Above \$1,000 (e.g., \$1,052.50)	Premium — bond's rate > market rate
Exactly 100	Exactly \$1,000	Par — bond's rate = market rate
Below 100 (e.g., 98.00)	Below \$1,000 (e.g., \$980.00)	Discount — bond's rate < market rate

Calculating Current Bond Yield

The **current yield** tells you what percentage return you earn annually based on the price you actually paid — not the par value.

Formula:

$$\text{Current Yield} = \text{Annual Interest} \div \text{Dollar Price of Bond}$$

Note that annual interest is always calculated on the **par value**, not the purchase price:

$$\text{Annual Interest} = \text{Stated Interest Rate} \times \text{Par Value (\$1,000)}$$

Worked Example — Meridian Advisory Group

Meridian purchased 5 bonds at a quoted price of 99.59. The stated interest rate is 4%.

Step	Calculation	Result
Dollar price per bond	$99.59\% \times \$1,000$	\$995.90
Total purchase cost (5 bonds)	$\$995.90 \times 5$	\$4,979.50
Annual interest per bond	$4\% \times \$1,000$	\$40.00
Total annual interest	$\$40.00 \times 5$	\$200.00
Current yield	$\$40.00 \div \995.90	4.02%

Notice that the current yield (4.02%) is slightly higher than the stated rate (4%) because the bond was purchased at a slight discount. When you pay less than par, every dollar of interest represents a higher percentage of your actual investment.

PART 3 · MUTUAL FUNDS & ETFs — POOLED INVESTING

The Case for Diversification

Consider a street vendor who sells both umbrellas and sunglasses. On rainy days, umbrella sales are strong. On sunny days, sunglass sales are strong. By carrying both, the vendor earns consistent revenue regardless of

the weather. This is the logic of **diversification** — spreading investments across assets that do not all move in the same direction at the same time.

Research suggests you need at least a dozen carefully selected individual stocks across multiple sectors to be meaningfully diversified. For a beginning investor, building and monitoring that portfolio takes considerable time, skill, and capital. That is where **pooled investment vehicles** come in.

Mutual Funds vs. ETFs

Both mutual funds and ETFs are "baskets" of investments — one purchase gives you exposure to many underlying securities. Their key differences:

Feature	Mutual Fund	ETF (Exchange-Traded Fund)
Pricing	Once per day, at market close	Real-time throughout the trading day
Trading	Priced and traded at end of day only	Trades like a stock — any time during market hours
Minimum Investment	Often \$500–\$3,000 minimum	As low as one share
Tax Efficiency	May distribute capital gains to all holders	More tax-efficient in taxable accounts via "in-kind" exchanges

One important warning: buying a single-sector ETF (technology only, for example) does not provide true diversification. If the entire tech sector declines, a tech-only ETF declines with it. True diversification requires exposure across multiple sectors — technology, healthcare, energy, financials, consumer goods, and more.

Net Asset Value (NAV)

Every mutual fund calculates its **Net Asset Value (NAV)** at the end of each trading day. NAV is the dollar value of one share of the fund.

Formula:

$$\text{NAV} = (\text{Total Market Value of Fund Holdings} - \text{Total Liabilities}) \div \text{Shares Outstanding}$$

Worked Example — Meridian Advisory Group Growth & Income Fund

Input	Value
Market value of fund holdings	\$5,550,000
Current liabilities	\$770,000
Shares outstanding	600,000

- Net assets: $\$5,550,000 - \$770,000 = \text{\$4,780,000}$
- NAV per share: $\$4,780,000 \div 600,000 = \text{\$7.97 per share}$

If an investor wanted to purchase 80 shares, they would pay: $\$7.97 \times 80 = \text{\$637.60}$.

FORMULA REFERENCE

Concept	Formula
Bid-Ask Spread	Ask Price – Bid Price
Dividend Yield	Annual Dividend ÷ Stock Price
Preferred Dividend (total)	Rate × Par Value × Preferred Shares
Common Dividend per Share	(Total Dividends – Preferred Dividends) ÷ Common Shares
Return on Investment (ROI)	(Net Proceeds – Total Buy Cost + Dividends) ÷ Total Buy Cost
Bond Dollar Price	Quoted % × Face Value (\$1,000)
Bond Current Yield	Annual Interest ÷ Dollar Price
Mutual Fund NAV	(Total Assets – Total Liabilities) ÷ Shares Outstanding
Budget Test	<code>=IF(Cost<=Budget,1,0)</code>
Income or Return Test	<code>=IF(OR(Yield>=MinYield,ROI>=MinROI),1,0)</code>
Risk-Proxy Test	<code>=IF(OR(Type="Bond",Type="Diversified mutual fund"),1,0)</code>
Overall Eligibility	<code>=IF(AND(BudgetTest=1,PerformanceTest=1,RiskTest=1),1,0)</code>

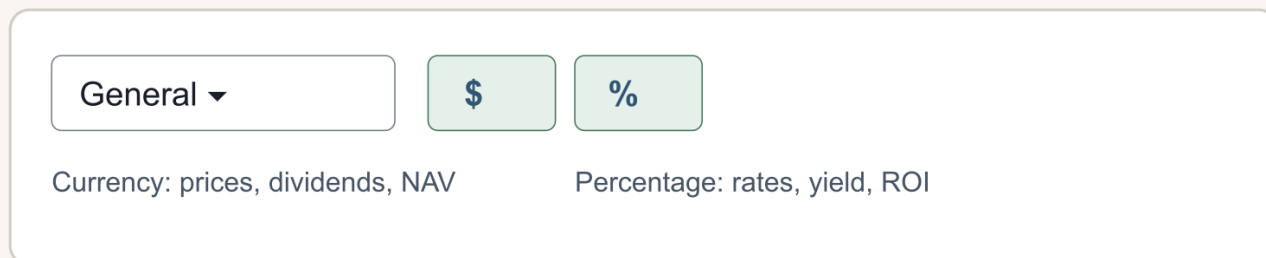
EXCEL SETUP: INPUTS FIRST, OUTPUTS SECOND

Open the starter workbook and select **Live You Try It**. Each practice block uses the same beginner-friendly model:

1. Column **B** names the business item.
2. Yellow cells in column **C** hold either a given input or your formula.
3. Column **D** gives self-check feedback; it is not an answer cell.
4. Column **E** provides a hint.

Use the **Name Box** to confirm the selected cell, then look at the **Formula Bar** to confirm what you typed. These screenshots show Windows Excel. Mac Excel includes the same core tools, but commands may appear in a different position.

Windows Excel · Home tab · Number group



Mac Excel has the same formats, but the controls may be positioned differently.

Build the Bond Model in Cells C51:C60

Select **C51** and enter 1000 . Select **C52** and enter 4% . Select **C53** and enter 99.59 — not 99.59% . Select **C54** and enter 5 .

Labeled worksheet model · Live You Try It

Row	B · Item	C · Student cell
51	Face value	1000
52	Stated interest rate	4%
53	Quoted price	99.59
54	Number of bonds	5
56	Dollar price per bond	=C53/100*C51
60	Current yield	=C58/C56

Then enter these formulas one cell at a time:

Select this cell	Type this exact formula	Format	Expected result
C56	=C53/100*C51	Currency, 2 decimals	\$995.90
C57	=C56*C54	Currency, 2 decimals	\$4,979.50
C58	=C52*C51	Currency, 2 decimals	\$40.00
C59	=C58*C54	Currency, 2 decimals	\$200.00
C60	=C58/C56	Percentage, 2 decimals	4.02%

Important distinction · bond quote vs. Excel percentage

Quoted price input

99.59

Type the market quote as a number.

Rate input

4%

Type the coupon rate as a percentage.

Formula: =C53/100*C51 · dividing the quote by 100 converts it to 99.59%.

Reasonableness checks: Because the quote is slightly below 100, the dollar price should be slightly below \$1,000. Because the bond sells below par, current yield should be slightly above the 4% stated rate. If C56 shows \$99,590 or \$9.96, recheck the `/100` conversion.

Automatic-recalculation test: Change **C53** from `99.59` to `98`. C56 should change to **\$980.00**, C57 to **\$4,900.00**, and C60 to about **4.08%**. Undo the change or type `99.59` again before continuing.

Read an ROI Formula from the Formula Bar

For the ROI practice block, enter the slide givens in **C33:C37**. Build the helper calculations in **C39:C45**. Finally, select **C46** and type `=(C44-C41+C45)/C41`, then apply Percentage with two decimals. The expected ROI is **6.12%**.

Formula bar · active cell C46

C46

`=(C44-C41+C45)/C41`

Read the references aloud:

net sale proceeds – total buy cost + dividends, all divided by total buy cost

Reasonableness check: the selling price is higher than the purchase price and dividends are positive, so ROI should be positive. It should not be enormous because commissions reduce the gain. Change **C37** from `1.21`

to 0 as a quick test; ROI should fall to about **3.06%**, proving the result is linked to the dividend input. Restore 1.21 afterward.

Class Challenge: Begin in Class, Finish for Homework

Keep **Class Challenge** separate from Live You Try It. The challenge is one connected Meridian client model rather than a list of isolated answers:

1. Build the linked calculations for the MAGH stock, Meridian bond, and Meridian Balanced Fund.
2. Link each result into the comparison table instead of retyping calculated values.
3. Test the \$8,000 budget, the income-or-return threshold, and the lesson-supported lower-risk proxy.
4. Select an eligible recommendation and explain the cost, performance evidence, risk proxy, and why the other alternatives fail.

Begin the model during class and finish it as homework. Use the **FormulaReferenceCard** for syntax reminders after you decide which metric or test belongs in a cell. This pre-reading does not provide the graded formulas, completed comparison table, or recommendation.

CHECK YOUR UNDERSTANDING

Answer these seven questions before class. Bring your work — these may be discussed.

1. Meridian recommends a stock currently trading at \$52.40 with a bid of \$52.33 and an ask of \$52.47. What is the bid-ask spread?
 2. A company has 500 shares of 6% preferred stock with a par value of \$100, and 2,000 shares of common stock. Total dividends declared: \$8,000. How much does each common shareholder receive per share?
 3. An analyst at Meridian bought 150 shares at \$28.50 and sold them one year later at \$31.00. Commission rate: 1% on both sides. No dividend was paid. What is the ROI?
 4. A bond is quoted at 103.75. What is the dollar price per bond? Is this a premium, discount, or par bond?
 5. A 5% bond is quoted at 97.50 (\$975.00 per bond). What is the current yield?
 6. Meridian's Balanced Fund has \$8,400,000 in holdings, \$950,000 in liabilities, and 1,200,000 shares outstanding. What is the NAV per share?
 7. An investor wants to buy 100 shares of the fund in Question 6. How much will they pay at NAV?
-

Answer Key

Q	Answer	Key calculation
1	\$0.14	$\$52.47 - \52.33
2	\$2.50/share	Preferred = $6\% \times \$100 \times 500 = \$3,000$; Remainder = $\$5,000$; $\$5,000 \div 2,000$
3	7.75%	Buy cost = $\$4,346.25$; Sell proceeds = $\$4,611.50$; ROI = $\$265.25 \div \$4,346.25$
4	\$1,037.50; Premium	$103.75\% \times \$1,000$
5	5.13%	Annual interest = $5\% \times \$1,000 = \50 ; $\$50 \div \975.00
6	\$6.21/share	$(\$8,400,000 - \$950,000) \div 1,200,000$
7	\$621.00	$\$6.21 \times 100$

KEY VOCABULARY

Term	Definition
Stock	A share of ownership in a company, entitling the holder to a portion of assets and profits
Common Stock	Stock that provides voting rights; paid dividends after preferred shareholders
Preferred Stock	Stock without voting rights but with priority on dividends and asset claims
Dividends in Arrears	Unpaid dividends owed to cumulative preferred shareholders that accumulate over time
Ticker Symbol	A unique letter code identifying a publicly traded security on an exchange
Bid-Ask Spread	The difference between the highest buyer price (bid) and lowest seller price (ask)
P/E Ratio	Price-to-Earnings ratio — stock price divided by earnings per share
Bond	A debt instrument; the investor lends money to a company in exchange for interest and repayment of principal
Par Value	The face value of a bond, typically \$1,000; the amount repaid at maturity
Current Yield	Annual bond interest divided by the bond's current dollar price
Net Asset Value (NAV)	The per-share value of a mutual fund's holdings after subtracting liabilities
Diversification	Spreading investments across different assets to reduce the impact of any single loss
ETF	Exchange-Traded Fund — a basket of securities that trades on an exchange like a stock